ReadVibe

AI-Powered Immersive Reading & Audio Conversion Platform

PROJECT OBJECTIVES

Formal Statement of Goals & Success Criteria

Final Year B.Tech Project — Information Technology
Anjalai Ammal Mahalingam Engineering College, Thiruvarur
Tamil Nadu, India | Academic Year 2025 – 2026

Project Objectives

ReadVibe is designed to solve a convergent set of problems across digital reading, accessibility, AI-assisted learning, and academic research. The following five objectives define the measurable goals of the project and serve as the primary success criteria for evaluation by the faculty panel and industry mentors.

01	02	03	04	05
Immersive Reading Modes	AI Content Intelligence	Multi-Tone Audio Engine	Research Paper Pipeline	Scalable Architecture

Objective 01

Deliver a Multi-Modal, Immersive Reading Experience

- Design and implement three distinct reading environments: Zen Mode, Reader Mode, and Vibe Reader Mode, each serving a different cognitive and sensory need of the user.
- Zen Mode provides a distraction-free focus environment where ambient tonal soundscapes (rain, forest, binaural beats — no lyrics) play beneath the document text displayed on a calming animated background.
- Vibe Reader Mode synchronizes the user's own music playlist (via Spotify or local library) with a dynamic AI-generated visual ambiance, displaying document text as an overlay in place of song lyrics.
- Reader Mode offers a clean, professional document viewer supporting PDF, EPUB, DOCX, and plain text formats with annotations, highlights, and accessibility-first typography options including dyslexia-friendly fonts.
- All three modes support offline access, customizable scroll speed, and seamless mode-switching without losing the user's reading position.

Key Technologies:

Flutter (Dart), Spotify Web API, Web Audio API, local file playback, animated canvas rendering

Objective 02

Implement AI-Powered Content Intelligence

- Integrate large language models (Claude API, GPT-4, Gemini) to perform intelligent operations on any user-uploaded document — including automatic summarization in brief, standard, and deep-dive formats.

- Generate concept explanation diagrams, block diagrams, and flowcharts directly from document content using Mermaid.js and GPT-4 Vision, exportable as PNG or embeddable HTML.
- Produce interactive mindmaps from chapter headings and key concepts, giving visual learners a structured overview of any document.
- Surface contextual reference images from open image APIs (Unsplash, Google Custom Search) to visually illustrate the concepts being read.
- Provide a "Chat with this Book" AI assistant powered by Retrieval-Augmented Generation (RAG), where the LLM answers user questions strictly grounded in the uploaded document's content using Pinecone vector embeddings.

Key Technologies:

Claude API, OpenAI GPT-4, Google Gemini, LangChain, Pinecone / FAISS, Mermaid.js, Unsplash API

Objective 03

Build a Flexible, Multi-Tone Text-to-Audio Conversion Engine

- Develop a full document-to-speech pipeline using ElevenLabs API and OpenAI TTS that converts any uploaded PDF, EPUB, or DOCX into a high-quality AI-narrated audiobook.
- Implement three distinct voice personality modes: Humor Tone (upbeat, expressive delivery suited for comic books and light fiction), Tough Tone (authoritative, serious narration suited for historical and academic texts), and Regular Tone (neutral, clear delivery for general-purpose reading).
- Support single-voice narration as well as a Collaborative Audiobook Studio where two or more users are assigned different chapters or character voices, recorded via WebRTC, and stitched into a unified audiobook with AI post-processing for noise reduction and normalization.
- Provide granular playback controls: adjustable speed (0.5× to 3×), chapter-level navigation, bookmarking, and a sleep timer; finished audiobooks are downloadable as MP3.
- Enable asynchronous audio generation via RabbitMQ message queue, so large documents are processed in the background while users continue reading.

Key Technologies:

ElevenLabs API, OpenAI TTS, WebRTC, RabbitMQ, FFmpeg (audio stitching), AWS S3 (audio storage)

Objective 04

Enable Research Paper Drafting & Academic Support Pipeline

- Provide research scholars and final-year students with a dedicated deep research pipeline: users upload source documents or enter a research topic, and the AI automatically scours academic databases (Semantic Scholar, Crossref) and the web for relevant papers, credible references, and citations.
- Present retrieved references in an organized, clickable list with relevance scores, allowing users to include or exclude specific sources before paper generation begins.
- Auto-draft a complete, structured research paper — Abstract, Introduction, Literature Review, Methodology, Results & Discussion, Conclusion, and References — through iterative conversational prompting, giving users full control over tone and depth at each stage.
- Support export to DOCX or PDF in IEEE and APA citation formats, making submissions ready for journals and conferences with minimal manual formatting effort.
- Integrate a plagiarism-awareness layer that cross-checks generated content and flags sections needing further paraphrasing or citation.

Key Technologies:

Semantic Scholar API, Crossref API, Claude API (long-context), python-docx, LaTeX (optional IEEE format), PostgreSQL

Objective 05

Ensure Scalability, Accessibility & Real-World Deployment Readiness

- Engineer the entire platform using a containerized microservices architecture — Auth, Document, Audio, AI, and Research services are independently deployable Docker containers orchestrated via Kubernetes on AWS EKS, enabling horizontal scaling under high user load.
- Design every API endpoint following RESTful standards with JWT-secured authentication, rate limiting, and comprehensive error handling, ensuring the backend is production-grade from day one.
- Prioritize accessibility throughout the mobile UI: support for screen readers (TalkBack / VoiceOver), dyslexia-friendly fonts (OpenDyslexic), high-contrast themes, adjustable text sizes, and one-handed navigation design.
- Implement offline capability for downloaded documents and pre-generated audio, allowing users in low-connectivity areas (rural students, field researchers) to access all core features without an active internet connection.
- Establish a CI/CD pipeline using GitHub Actions and AWS CodeDeploy for automated testing, linting, and deployment to production, ensuring the project is maintainable and extensible beyond the academic submission.

Key Technologies:

Docker, Kubernetes (AWS EKS), GitHub Actions, AWS CodeDeploy, Firebase Auth, Redis (caching), Flutter offline storage

Objectives at a Glance

#	Objective	Core Outcome	Target Users
01	**Immersive Reading Modes**	Three distinct mode environments (Zen, Vibe, Reader) offering tailored focus and engagement.	*Students, Casual Readers*
02	**AI Content Intelligence**	Automatic summarization, diagram generation, mindmaps, reference images, and document Q&A.	*All Users, Researchers*
03	**Multi-Tone Audio Engine**	High-quality TTS with Humor / Tough / Regular tones + collaborative multi-speaker audiobook studio.	*Learners, Book Clubs*
04	**Research Paper Pipeline**	End-to-end research workflow: source upload → reference discovery → auto-drafted paper export.	*Research Scholars, PG Students*
05	**Scalable Deployment**	Microservices architecture, CI/CD pipeline, accessibility features, and offline-first mobile design.	*All Users, Enterprise*